

SupplyFlow Supply Chain Data Pipeline

S3 to Redshift Snowflake Data Warehouse with Power BI Analytics

SupplyFlow is a serverless AWS-style data engineering project that transforms supply-chain and geospatial delivery data into a normalized analytical warehouse. The project uses S3 for staging, Lambda for orchestration, Glue PySpark for ETL, Redshift for warehousing, and Power BI for business intelligence.

Project Aim and Scope

The goal is to demonstrate a reproducible cloud data pipeline that ingests operational CSV and route-style geospatial data, builds an eight-dimension and one-fact snowflake schema, and supports profitability, carrier performance, and delivery-risk reporting.

Repository Organization

Folder	Purpose
infrastructure/	IAM and Redshift/VPC reference configuration
source_data/	Raw CSV, GeoJSON, and metadata source files
etl_jobs/	Lambda and Glue ETL scripts
data_warehouse/	Redshift DDL for the snowflake schema
bi_dashboards/	Power BI specification and dashboard preview
documentation/	Final project report
image/	Architecture diagram assets

Data Warehouse Model

Table	Analytical Role
dim_date	Calendar attributes for order, ship, promised, and actual dates
dim_region	Normalized geography hierarchy
dim_location	City, state, region, latitude, and longitude
dim_customer	Customer segment and priority
dim_product_category	Product category hierarchy
dim_product	Product price, cost, and weight attributes
dim_carrier	Carrier, mode, tier, and cost basis
dim_service_level	Standard, expedited, and same-day SLA definitions
fact_shipments	Shipment measures, profitability, late days, and risk score

ETL Transformation Logic

The Glue job joins orders, shipments, carriers, customers, products, and locations; generates date and dimensional keys; calculates gross profit, profit margin, delivery days, late days, on-time flag, damage flag, and delivery risk score; and writes warehouse tables to S3 or Redshift.

AWS Services

Service	Function
Amazon S3	Stores raw data and curated warehouse outputs
AWS Lambda	Routes S3 events and processes optional GeoJSON files
AWS Glue	Runs PySpark ETL and schema discovery crawlers
Amazon Athena	Verifies cataloged data before loading
Amazon Redshift	Hosts the snowflake schema for analytics
CloudWatch	Captures operational logs
Power BI	Presents executive, carrier, and delivery-risk dashboards

Business Intelligence Outcomes

The dashboard design includes an Executive Overview for revenue and margin, Carrier Performance for reliability and cost analysis, and Delivery Risk for regional hotspots and exception drill-through.

End Result

The finished project is a student-safe, local-first AWS portfolio project with deployable artifacts for S3, Lambda, Glue, Redshift, and Power BI. It demonstrates cloud architecture and data modeling without requiring paid cloud resources for validation.